JOHNS HOPKINS UNIVERSITY
WHITING SCHOOL OF ENGINEERING
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Propeller-Assisted Robust 3D Hopping Robot with Hierarchical Force Allocation

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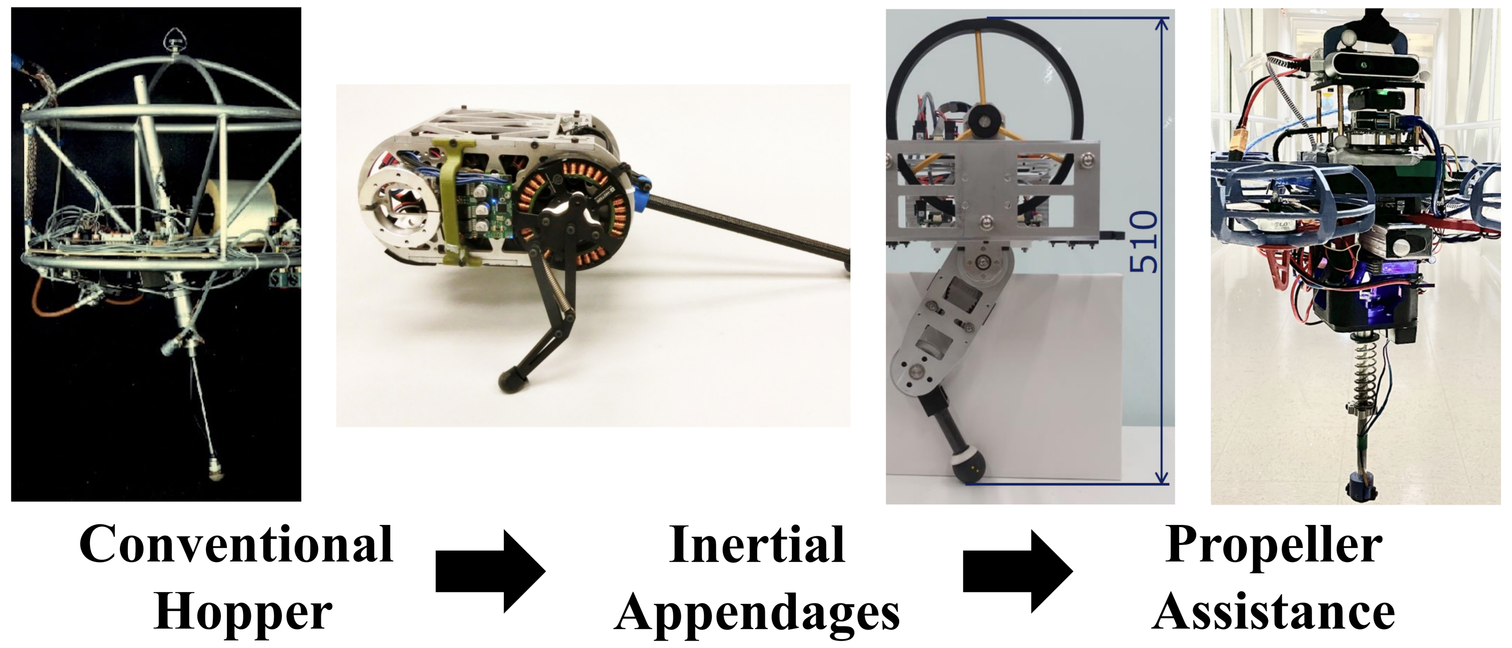
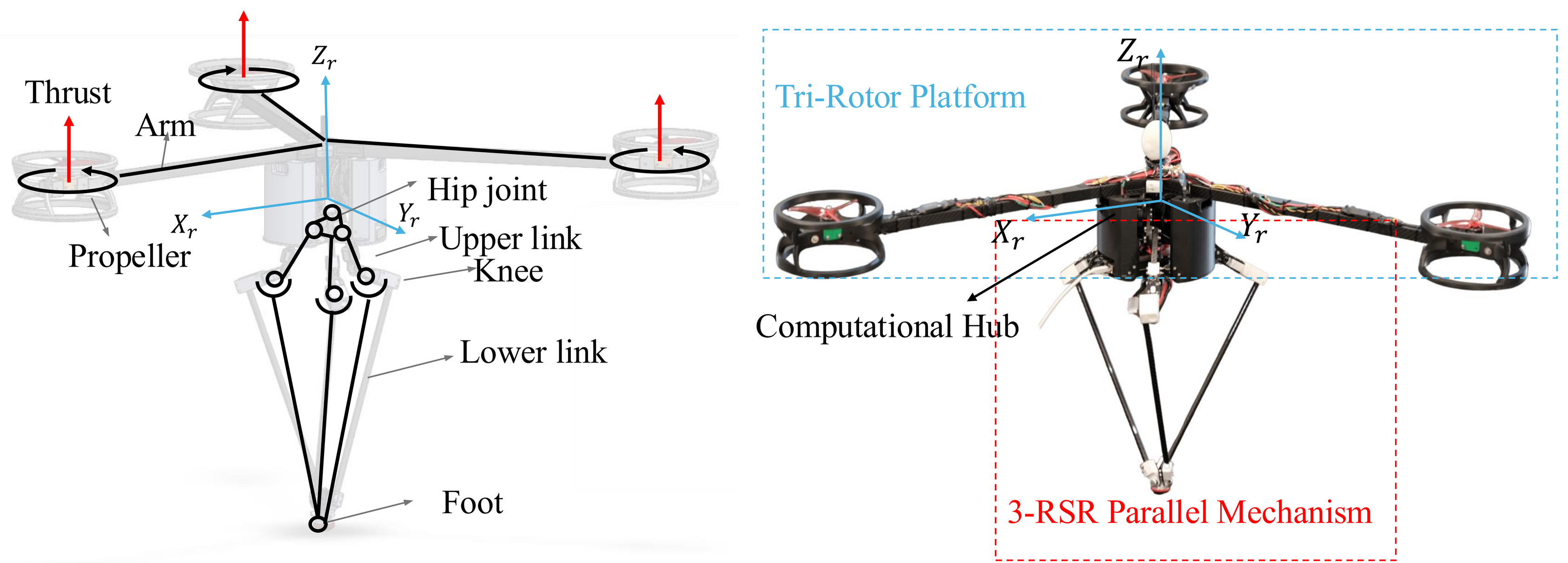
ACCEPTED - IEEE CASE 2026

1. Problem - no contact authority in flight



Tail-assisted, rotor-assisted, and coupled leg-rotor robots all target the same gap: contact authority disappears in flight.

2. Platform - leg first, rotors auxiliary

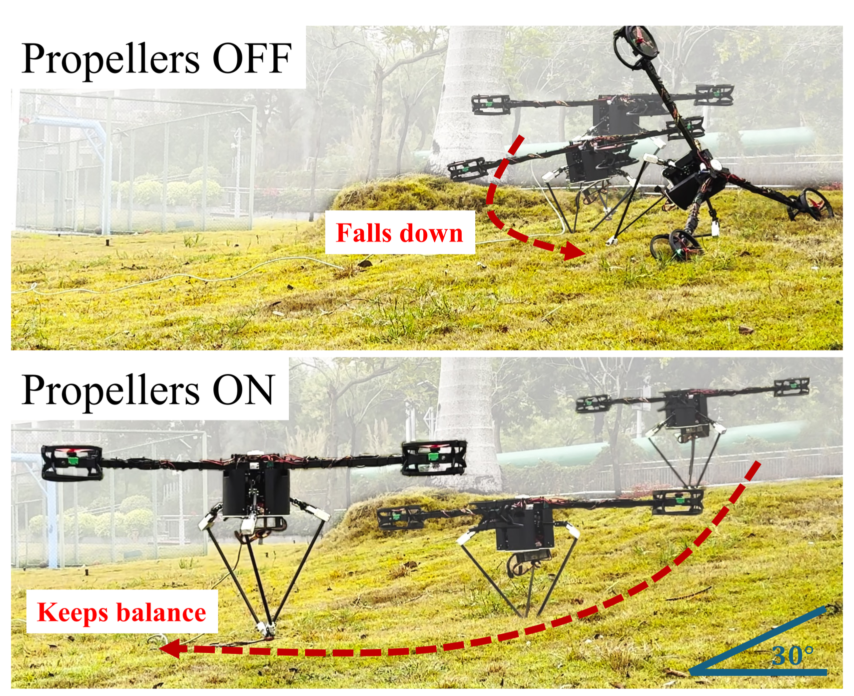


An active **3-RSR parallel leg** provides fast 3-DoF foot placement. A trunk-mounted **tri-rotor** supplies continuous roll/pitch authority while total thrust-to-weight stays below one.

3.75 kg total ▪ **0.35 kg leg** ▪ **20 Nm / motor** ▪ **500 Hz control**

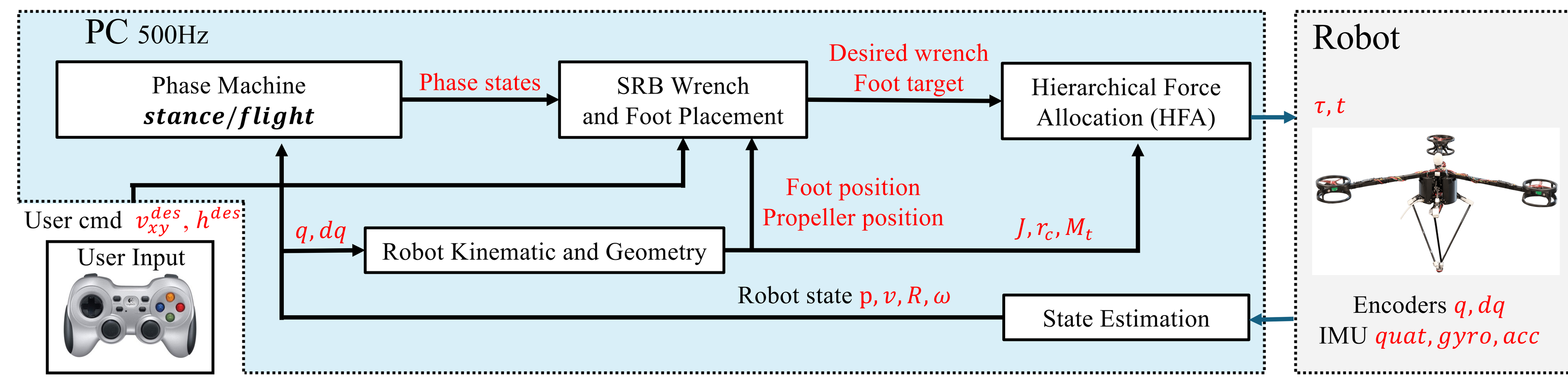
Light moving leg + rigidly mounted rotor assembly makes the SRB approximation useful while preserving fast omnidirectional foot placement.

3. First proof - stable on 30° grass



Propellers off: attitude diverges and the robot falls. **Propellers on:** residual roll/pitch moment remains available in flight, so the robot keeps hopping upright.

4. Implementation - real-time control

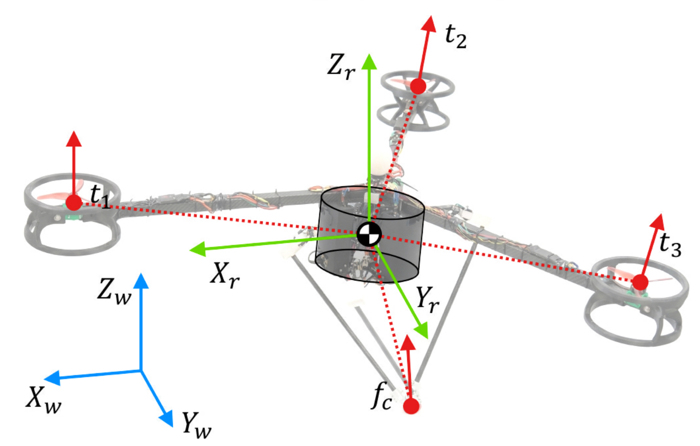


The PC estimates state, plans the body wrench, and runs HFA. The Raspberry Pi sends leg torque and rotor thrust commands.

LCM / Ethernet latency below 0.5 ms for 99% of 500 Hz cycles

Sense encoders + IMU → **Estimate** p, v, R, ω → **Allocate** leg torque + rotor thrust

5. Method - leg-priority HFA



Paper Fig. 4 - SRB actuation map. Stance contact force $\boldsymbol{f}_c$ and rotor thrusts t_i jointly generate the body wrench.

1. LEG-PRIORITY STANCE

$$\boldsymbol{f}_{c,xy}^{cmd} = \arg\min_{\boldsymbol{f}_{xy}} \|(\boldsymbol{r}_c \times \boldsymbol{f}_c)_{rp} - (\boldsymbol{\tau}^{des})_{rp}\|_2^2$$

$$\|\boldsymbol{f}_{xy}\|_2 \le \mu f_{c,z}, \quad \|\boldsymbol{J}^\top(-\boldsymbol{R}^\top \boldsymbol{f}_c)\|_\infty \le \tau_{max}$$

2. PROPELLER RESIDUAL

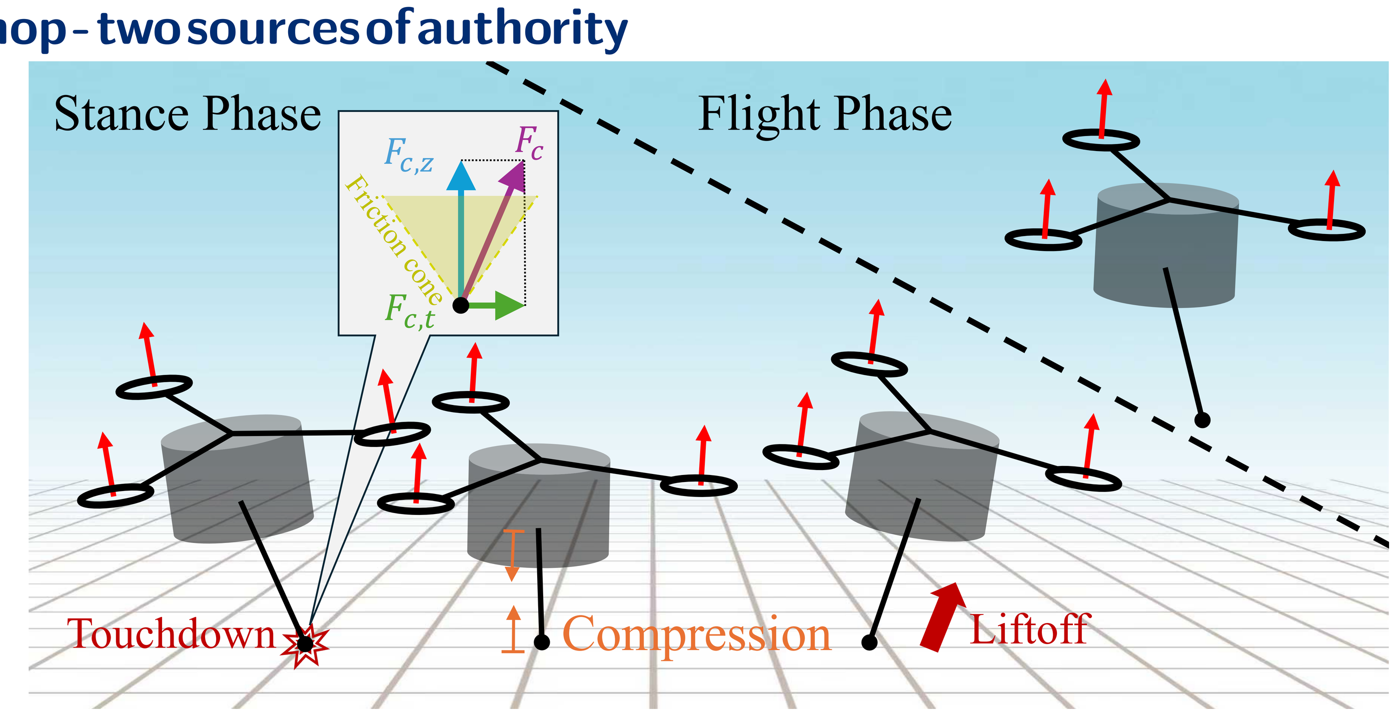
$$\boldsymbol{\tau}_{res}^{cmd} = \boldsymbol{\tau}^{des} - \boldsymbol{r}_c \times \boldsymbol{f}_c^{cmd}$$

Rotors receive only the moment left after the feasible leg allocation.

3. FLIGHT: $\boldsymbol{f}_c = 0$; rotors become the primary attitude actuator while the leg tracks $\boldsymbol{r}_{td,xy}^{des} = k_v \boldsymbol{v}_{xy} - k_r \boldsymbol{v}_{xy}^{des}$.

Leg first in stance; rotor residual in stance; rotor attitude authority throughout flight.

6. One hop - two sources of authority



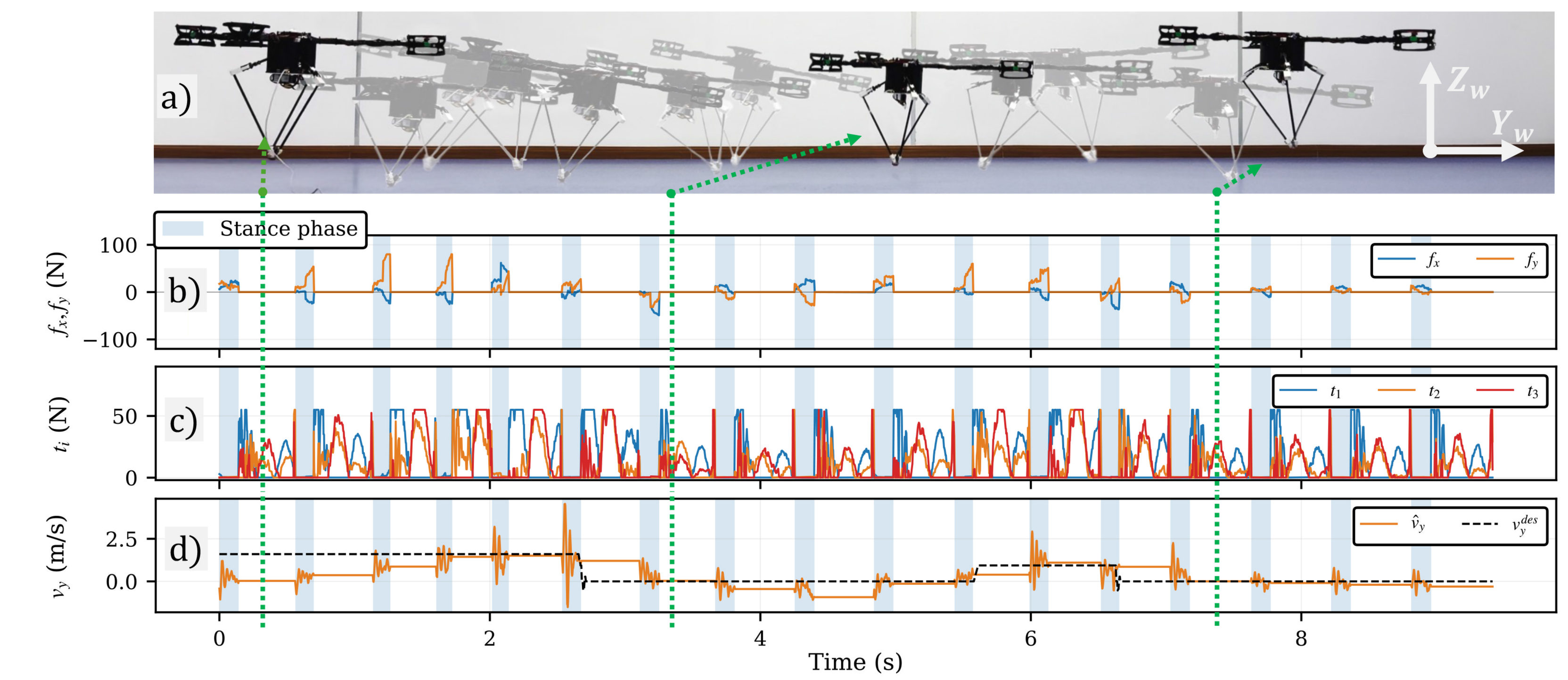
1. TOUCHDOWN
$l \le l_0 - \Delta_{td}$ enters stance. The leg absorbs impact, then the energy law commands vertical force to reach the next apex.

2. STANCE ALLOCATION
The leg realizes feasible force and moment under friction and motor limits. Rotors supply only the residual roll/pitch moment.

3. LIFTOFF + FLIGHT
$l \ge l_0$ removes the contact wrench. Rotors become the primary attitude actuator while the leg moves to the next Raibert foot target.

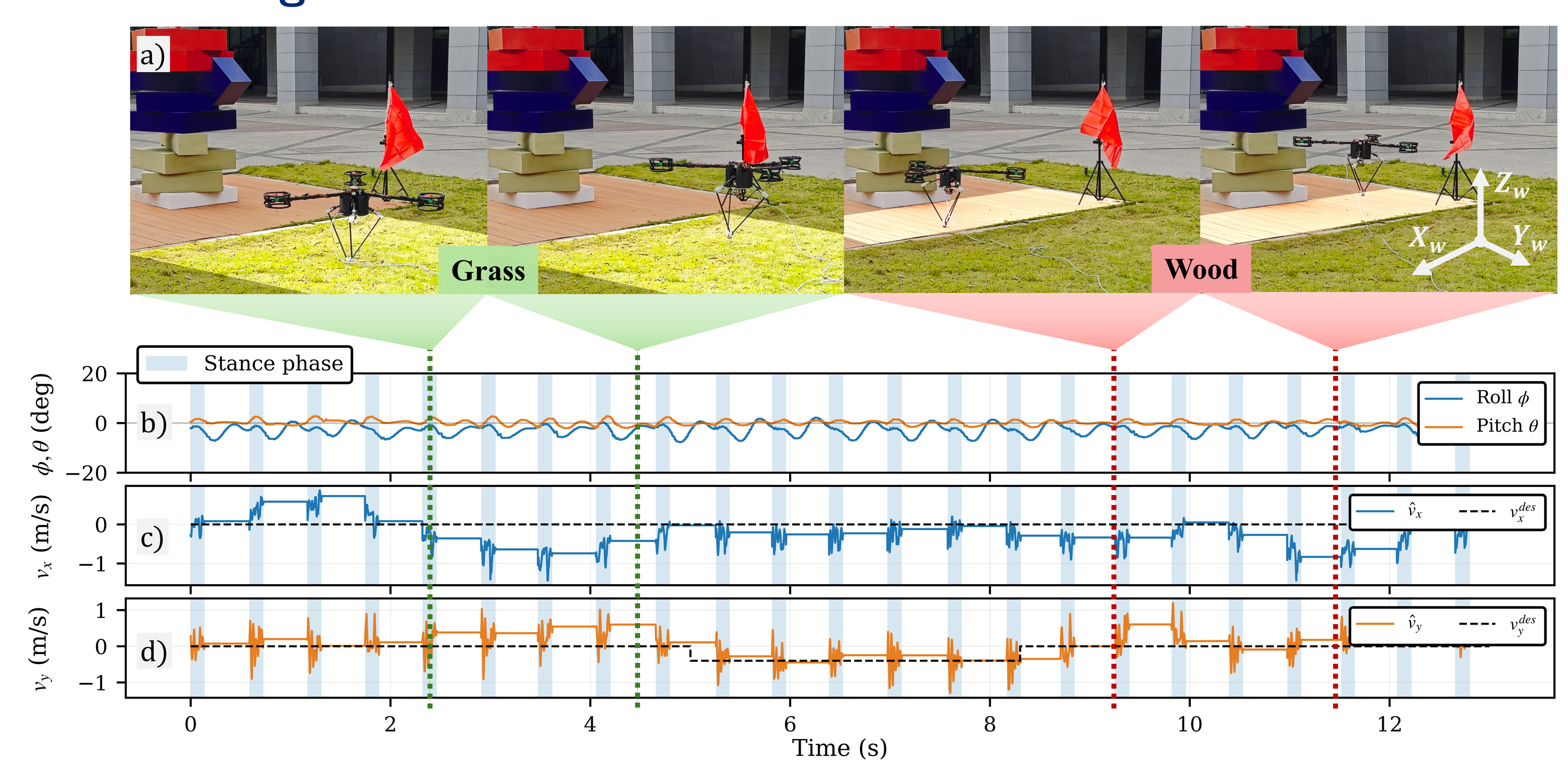
Event-driven handoff: leg authority exists only in stance; rotor attitude authority remains continuous through the full hop.

7. Validation - indoor tracking



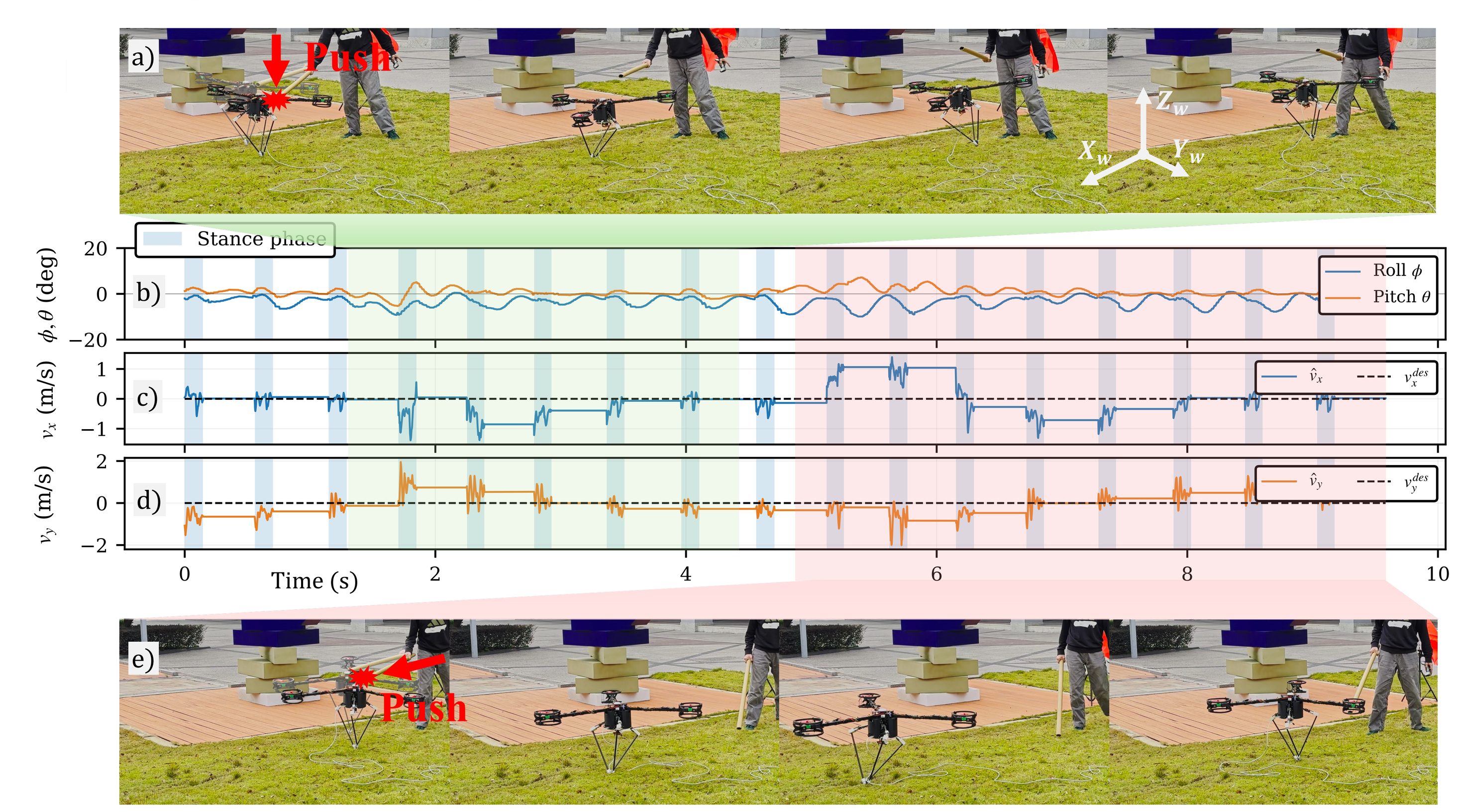
17 hops - 1.6 and 0.94 m/s commands - bounded roll and pitch through every impact.

8. Validation - grass to wood



23 hops - same controller - no terrain model - no gain retuning.

9. Validation - push recovery



Rotor-arm hit: attitude recovers in 1-2 hops. Main-body hit: lateral velocity recovers in three hops.